

Interpretation of Results

The code execution provided the following results based on the scenario:

Number of Nodes 5

Number of Edges 5

Degree Distribution {'PA': 3, 'DO': 3, 'ES': 2, 'MO': 1, 'AT': 1}

Isolated Nodes []

Number of Nodes and Edges

- Number of Nodes (): The network consists of five researchers, representing the total size of the research group and its immediate collaborators for this work conducted.
- Number of Edges (): There are five unique, defined connections or collaboration links amongst us. This indicates the network is relatively sparse; for a complete graph with N=5, the maximum number of possible edges is $N(N-1)/2 = 10$

Degree of Distributionh

The degree of a node is the number of connections it has.

Node Degree Interpretation

PA	3	High Centrality: Dr. Prince Ansah acts as a central hub, connecting the two MSc students (ES, DO) and the external collaborator (AT).
DO	3	High Centrality (Active Collaborator): David Ofosu (DO) is also a central hub, connected to his supervisor (PA), his peer (ES), and an external collaborator (MO). This shows he is highly active in both internal and external collaborations.
ES	2	Core Member: Esther Sevor (ES) is connected to her supervisor (PA), and her peer (DO). She is a core member, but her network ties are limited to the immediate UG research environment.
AT	1	Peripheral Collaborator: (AT) is only linked to the main faculty hub (PA). This suggests a specific, focused collaboration without broader internal group involvement.

MO 1 Peripheral Collaborator: Mary O'Reilly (MO) is only linked to the active student (DO) through the visiting collaboration.

Overall Interpretation: The distribution is non-uniform, identifying two clear hubs (PA and DO) who drive most of the connections and act as gatekeepers or bridges for external collaboration (MO and AT).

C. Isolated Nodes

- Isolated Nodes: [] (None)

Interpretation: There are no isolated nodes. Every researcher is connected to at least one other member, ensuring that all five individuals listed in the scenario were actively involved in some aspect of the underlying research and are part of the single cohesive network component.